A Market in This Room

ECON 201: Principles of Microeconomics
Lecture 14

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So Far: Firms That Set Prices

- We have studied how firms set prices: a car company selling a specialty car, a brand of cereal like Cheerios, AT&T and T-Mobile picking prices strategically.
- Each sold a differentiated product. Nothing else on the market was a perfect substitute for it.
- That gave them **market power**: they could charge above marginal cost, because no rival could undercut them and take all their customers.
- Wanda kept her loyal customers at the high price, and how much Cheerios could mark up depended on how elastic its demand was.

Markets Where No One Picks the Price

- Many sellers have no such power. Think of three taco stands at a Saturday flea market: the same tacos, no brand, a few steps apart, so buyers walk around and compare. Is any one of them picking the price?
- Where goods are hard to tell apart and there are many sellers, firms take the price as given: used textbooks, raw cotton, produce at a farmers market.
- Where does that price come from, and where does it settle? Before the model, we build such a market here.

The Game

- Half of you are buyers and half are sellers of half a dozen tamales. Each of you gets one card; keep its number private.
- **Buyers**: the number on your card is your **value**, the most the tamales are worth to you. Find a seller who will sell for less.

$$\text{Buyer's gain} = \text{value} - \text{price}$$

- **Sellers**: the number on your card is your **cost**. Find a buyer who will pay more.

$$\text{Seller's gain} = \text{price} - \text{cost}$$

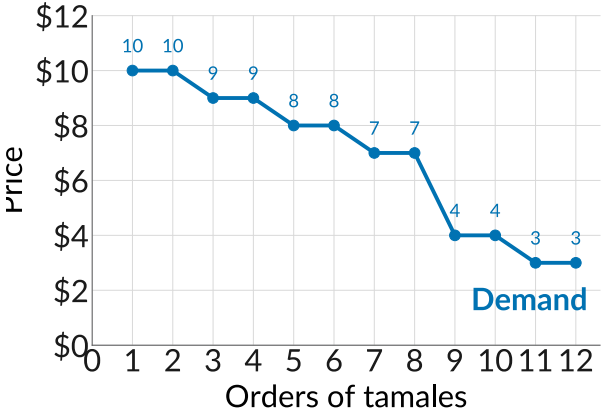
- When a pair agrees, come to the front with your cards and I record it. One trade each per round, never at a loss. Five minutes a round, then we reshuffle.

What Happened to Prices?

- Where did the Round 1 prices land, and how spread out were they?
- Did the prices settle as the rounds went on? Around what number?
- How many trades happened in each round? Who could not find a trade, and why?

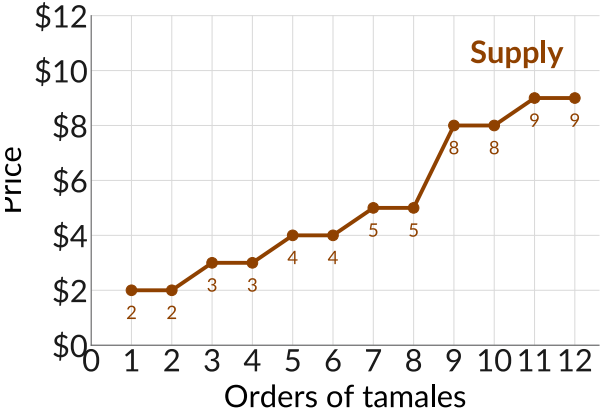
From Cards to a Demand Curve

- Say 12 buyers and 12 sellers hold our cards. The buyer cards, highest first: 10, 10, 9, 9, 8, 8, 7, 7, 4, 4, 3, 3.
- At $8, who buys? Everyone at 8 or above: six buyers. At $4, ten.
- Lined up this way, the cards are the **demand curve**: at each price, how many want to buy.



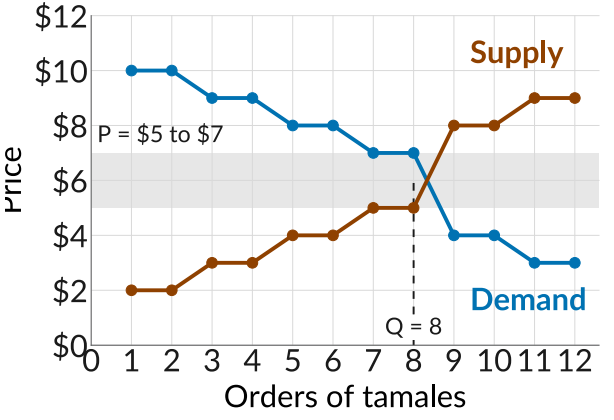
From Cards to a Supply Curve

- The twelve seller cards, lowest first: 2, 2, 3, 3, 4, 4, 5, 5, 8, 8, 9, 9.
- At $8, who sells? Everyone at 8 or below: ten sellers. At $4, six.
- Lined up this way, the cards are the **supply curve**: at each price, how many are willing to sell.



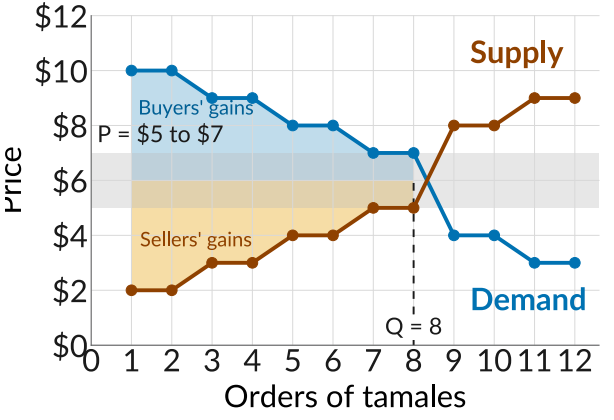
Why the Price Ends Up Where They Cross

- At $8: ten sellers, six buyers. Four sellers find no one and cut their price. That is **excess supply**.
- At $4: ten buyers, six sellers. Buyers with no seller offer more. That is **excess demand**.
- From $5 to $7 the counts match, eight and eight, and no one has a reason to move. That is the **equilibrium price** and **quantity**, and where your prices settled.



Who Trades and Who Does Not

- The eighth order pairs a cost of 5 with a value of 7. At $6 each gains $1, so it happens.
- The ninth would pair a value of 4 with a cost of 8. No price suits both. The market leaves out exactly the trades that lose value.
- The gains on the eight orders add up to $2 \times (8 + 6 + 4 + 2) = 40$, the most possible. The shading is that **surplus**: at $6, $20 to buyers and $20 to sellers.



Price-Takers

- A seller who asked more than the going price lost the sale to another seller, and a buyer who offered less found no seller. Each of you could only agree or walk away.
- Buyers and sellers who cannot gain by trading at any price other than the going price are **price-takers**. No one can move the price, but anyone can trade at it.
- A market where the quantity supplied equals the quantity demanded at the going price, and everyone is a price-taker, is in **competitive equilibrium**. That is where the later rounds ended up.